

Short Report: AO Unit Step Size Analysis

Experiment Setup:

- CMOS sensor: 62MP with 9576 x 6388 pixels [from datasheet of ZWO ASI 6200MM-PRO]
- Single pixel size: 3.76 μm
- AO unit: Starlight Xpress Active Optics Guiding System without OAG

Methodology:

We measured the pixel position of a reference point on the CMOS sensor after moving the AO unit in different directions (West, North, South, East) and returning to the center position each time.

Results and Calculations:

1. Center to West movement:

Start: (4169.111, 2641.752)

End: (4168.837, 2646.532)

$\Delta x = -0.274$ pixels, $\Delta y = 4.78$ pixels

Movement: $\sqrt{((-0.274 * 3.76)^2 + (4.78 * 3.76)^2)} = 18.00 \mu\text{m}$

2. West to Center:

Start: (4168.837, 2646.532)

End: (4170.048, 2647.622)

$\Delta x = 1.211$ pixels, $\Delta y = 1.09$ pixels

Movement: $\sqrt{((1.211 * 3.76)^2 + (1.09 * 3.76)^2)} = 6.13 \mu\text{m}$

3. Center to North:

Start: (4170.048, 2647.622)

End: (4169.844, 2638.253)

$\Delta x = -0.204$ pixels, $\Delta y = -9.369$ pixels

Movement: $\sqrt{((-0.204 * 3.76)^2 + (-9.369 * 3.76)^2)} = 35.24 \mu\text{m}$

4. North to Center:

Start: (4169.844, 2638.253)

End: (4169.567, 2626.795)

$\Delta x = -0.277$ pixels, $\Delta y = -11.458$ pixels

Movement: $\sqrt{((-0.277 * 3.76)^2 + (-11.458 * 3.76)^2)} = 43.09 \mu\text{m}$

5. Center to South:

Start: (4169.567, 2626.795)

End: (4166.804, 2615.408)

$\Delta x = -2.763$ pixels, $\Delta y = -11.387$ pixels

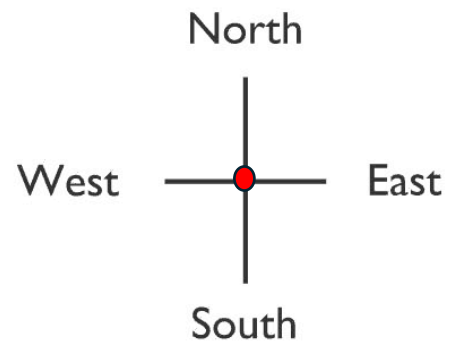
Movement: $\sqrt{((-2.763 * 3.76)^2 + (-11.387 * 3.76)^2)} = 44.22 \mu\text{m}$

6. South to Center:

Start: (4166.804, 2615.408)

End: (4166.321, 2619.758)

$\Delta x = -0.483$ pixels, $\Delta y = 4.35$ pixels



Movement: $\sqrt{((-0.483 * 3.76)^2 + (4.35 * 3.76)^2)} = 16.46 \mu\text{m}$

7. Center to East:

Start: (4166.321, 2619.758)

End: (4166.544, 2618.544)

$\Delta x = 0.223$ pixels, $\Delta y = -1.214$ pixels

Movement: $\sqrt{((0.223 * 3.76)^2 + (-1.214 * 3.76)^2)} = 4.64 \mu\text{m}$

8. East to Center:

Start: (4166.544, 2618.544)

End: (4166.321, 2618.926)

$\Delta x = -0.223$ pixels, $\Delta y = 0.382$ pixels

Movement: $\sqrt{((-0.223 * 3.76)^2 + (0.382 * 3.76)^2)} = 1.66 \mu\text{m}$

Key Findings:

Single Step Movements:

- West: 18.00 μm
- North: 35.24 μm
- South: 44.22 μm
- East: 4.64 μm

Maximum Theoretical Movement:

Given the maximum of 45 steps, we can estimate the maximum theoretical movement in each direction:

- West: $45 * 18.00 \mu\text{m} = 810 \mu\text{m}$
- North: $45 * 35.24 \mu\text{m} = 1,585.8 \mu\text{m}$
- South: $45 * 44.22 \mu\text{m} = 1,989.9 \mu\text{m}$
- East: $45 * 4.64 \mu\text{m} = 208.8 \mu\text{m}$

Asymmetry in Movement:

The AO unit shows significant asymmetry in its step sizes:

- North-South axis has larger movements (35-44 μm per step)
- East-West axis has smaller movements (4-18 μm per step)
- Variability in Return-to-Center:

The return movements to the center position often didn't match the original position exactly, indicating some hysteresis or inconsistency in the AO unit's performance.

Observations:

1. The Starlight Xpress Active Optics Guiding System demonstrates variable step sizes depending on the direction of movement.
2. The system has higher sensitivity in the North-South axis compared to the East-West axis.
3. With a maximum of 45 steps, the AO unit can theoretically provide a correction range of about 2 mm in the most sensitive direction (South) and about 200 μm in the least sensitive direction (East).
4. The asymmetry in step sizes might be a design feature to provide finer control in certain directions, or it could be due to the mounting orientation of the AO unit in the optical system.

5. The inconsistency in return-to-center movements suggests that the system may benefit from calibration or may have some inherent limitations in repeatability.

Conclusions:

The Starlight Xpress Active Optics Guiding System demonstrates variable step sizes depending on the direction of movement. The system appears to have higher sensitivity in the North-South axis compared to the East-West axis. The inconsistency in return-to-center movements suggests that the system may benefit from calibration or may have some inherent limitations in repeatability.

Further investigation might be needed to determine the cause of the center position drift and to establish a more consistent relationship between AO unit commands and actual image movement on the CMOS sensor.